

Cd-S doped glass Shields: Structural, optical, and nuclear shielding characteristics

Emran Eisa Saleh^{a,*}, Mohammed A. Algradee^{b,c}, Abdullah M. Qaid^{b,c}, Dua'a Anis Taya^a

^a Physics Department, Faculty of Science, University of Aden, P. O. 6312, Yemen

^b Department of Physics, Faculty of Science, Ibb University, 70270, Ibb, Yemen

^c Faculty of Science and Engineering, Aljazeera University, Ibb, Yemen

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ABSTRACT

This research explores the structural, optical, and nuclear shielding attributes of Cd-S doped glass shields. A series of five glass samples were synthesized with compositions specified as $5\text{Li}_2\text{O} + 33\text{NaF} + 7\text{ZnO} + 55\text{P}_2\text{O}_5 + \text{xwt} \% (\text{CdO} + \text{S})$, where $\text{x} = 0.0, 1.0, 2.0, 3.0$ and 4.0 . X-ray diffraction analysis confirmed the amorphous nature of the samples. Density measurements were conducted using the Archimedes method. The optical properties of the prepared glass samples were measured using UV-VIS spectrophotometer within the spectral range of 190–1100 nm. With an increase in (CdO + S) content from 0.0 to 4 wt%, the band gap energy (E_g) and Urbach energy (E_u) decreased from 4.12 eV to 3.14 eV and from 4.77 eV to 4.05 eV, respectively. Nuclear shielding parameters, including mass attenuation coefficient (MAC), linear attenuation coefficient (LAC), mean free path (MFP), half value layer (HVL), effective atomic number (Z_{eff}), electron density (N_{eff}), atomic and electron cross sections, exposure buildup factor (EBF), and energy absorption buildup factor (EABF), were computed using Phy-X PSD software across an energy range from 0.015 KeV to 15 MeV. The results revealed MAC values ranging from 9.37 to 15.34 cm^2/g , LAC values from 23.56 to 39.78 cm^{-1} , HVL values from 0.029 to 0.017 cm, MFP values from 0.04 to 0.025 cm, Z_{eff} values from 16.57 to 16.03, and N_{eff} values from $(4.83\text{--}3.06) \times 10^{23}$ electron/g. Additionally, neutron removal cross sections were determined and compared with commercial glasses and elements typically employed for radiation shielding. Sample with a higher weight percentage of (Cd + S), exhibited superior nuclear radiation shielding effectiveness based on the computed parameters.

1. Introduction

Glass is commonly utilized for shielding against gamma and X-rays due to its advantageous properties (Taya et al., 2023; Algradee et al., 2022a; Saleh et al., 2022). The optical properties of glass and its high transparency make it a desirable material for shielding. The combination of multiple types of glass with added heavy metal oxides makes it an excellent option for shielding against high-energy radiation. Furthermore, the properties of glass, such as simple preparation, low cost, and easily accessible, make it a suitable materials for radiation shielding. Glass's transparency makes it a convenient choice for preparing radiopharmaceuticals in medical imaging for nuclear medicine procedures (Saleh et al., 2022; Al-Fakeh et al., 2023; Murshed et al., 2024; Algradee et al., 2022b).

Oxyfluoride glasses have received great attention because it has characterized by extensive composition, low nonlinear refractive index

and high Abbe index and high content of doped semiconductor materials. They are promising host matrix for applications in optics. Also, their study provides basic information about glass forming ability (Algradee et al., 2016, 2021a; Saleh et al., 2021a, 2021b). The presence of F^- anions in these glasses influences their structures and properties by forming polyhedra with metal-fluorine bonds and facilitating the creation of cation-anion-cation bridges by monovalent fluorine ions. This leads to increase in viscosity near the glass transition temperature, crucial for glass formation. Furthermore, the inclusion of alkali fluorides in the glass composition can enhance their chemical resistance and thermal stability while reducing their hygroscopic behavior, thereby expanding their potential applications. Additionally, the high field intensity of alkali fluoride cations allow them to consolidate the glass networks through electrostatic attraction with F^- or O^{2-} anions, contributing to the prevention of phase separation. (Hager, 2008; Algradee et al., 2021b; Abduelwhab et al., 2013; Brekhovskikh et al.,

* Corresponding author.

E-mail addresses: eesas2009@yahoo.com (E.E. Saleh), algradi772001@yahoo.com (M.A. Algradee), edu-aa.anees.scie@aden-univ.net (D.A. Taya).

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2010).

Evaluating the radiation shielding capacity of glass in all its form is essential to assessing its effectiveness in radiation protection. To understand the ability of glass to protect against ionizing radiation, necessary measurements must be performed, such as measuring the nuclear shielding coefficients. Knowing the values of the nuclear shielding parameters such as mass attenuation coefficient, linear attenuation coefficient, and effective atomic number of glass enables us to optimize glass compositions for use in nuclear applications such as medical imaging, laboratories, and nuclear reactors to protect personnel and equipment from the dangers of ionizing radiation. The nuclear shielding ability of glass is enhanced by adding heavy metal oxides such as bismuth subsalicylate and lead oxides. In addition to improving the nuclear shielding ability of glass, the glass must be transparent and non-opaque, and these oxides must be harmless to the surrounding environment (EzhilPavaiP, 2017; Aboutaleb and Safi, 2016; (Mofreh Al-Esaei et al., 2024); Al-Esaei et al., 2024).

The addition of metals to the glass matrix introduces enhanced gamma ray attenuation features, making the doped glass more efficient in attenuating gamma rays and providing increased protection against radiation exposure. Through gamma rays attenuation studies, the specific mechanisms of gamma ray interaction with the doped glass structure can be elucidated, offering insights into the material's effectiveness as a shielding barrier for gamma radiation in various applications, including nuclear medicine, radiation therapy, and industrial radiation protection. Also, the advancements in glass technology have played a crucial role in enhancing radiation protection for operators working with radiological gloveboxes, providing a transparent and effective barrier against harmful radiation exposure in radiological environments (Al-Esaei et al., 2024).

The primary objective of this study is to investigate and analyze the structural, optical, and nuclear shielding properties of Cd-S doped glass shields. The research involves the synthesis of a series of five glass samples with varied compositions containing $5\text{Li}_2\text{O}+33\text{NaF}+7\text{ZnO}+55\text{P}_2\text{O}_5+\text{xwt}\%(\text{Cd}+\text{S})$, where $x = 0.0, 1.0, 2.0, 3.0$ and 4.0 . The amorphous nature of the samples was verified through X-ray diffraction analysis. Furthermore, density measurements were carried out utilizing the Archimedes method to ascertain the physical characteristics of the synthesized glass samples.

2. Materials and methods

2.1. Samples preparation

The prepared glass samples composed according formula $5\text{Li}_2\text{O}-33\text{NaF}-7\text{ZnO}-55\text{P}_2\text{O}_5$ with weight percentages of $x(\text{CdO}+\text{S})$ at $x = 0.0, 1.0, 2.0, 3.0$, and 4.0 wt%, using the melt quenching method. High-purity of starting materials of Li_2CO_3 , NaF , ZnO , P_2O_5 , CdO , and S were used as raw materials. The $\text{CdO}:\text{S}$ ratio was maintained at 1:1 in all compositions. Sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) was added to create a reducing atmosphere during melting. The sucrose served to react with dissolved oxygen, preventing the removal of sulfur for the formation of cadmium sulfide particles. The chemical mixtures were placed in silica crucibles and sequentially heat-treated in an electric furnace at 100°C for 30 min, 250°C for 30 min, and 400°C for 15 min. This step prevents sulfur evaporation and contributes to the decomposition of P_2O_5 . The mixtures were then melted in a second furnace at 1100°C for 6 min and then rapidly quenched on a stainless steel plate to ensure uniformity. The molten glass samples were annealed at 250°C before gradually cooling to room temperature. To maintain integrity, the glass samples were stored in a desiccator to prevent moisture-induced damage. The glass samples are categorized based on the $(\text{CdO}+\text{S})$ wt% as follows.

- A: $\text{Li}_2\text{O}-\text{Na}-\text{Zn}-\text{P}-\text{Cd}-\text{S}$ (0.000)
- B: $\text{Li}_2\text{O}-\text{Na}-\text{Zn}-\text{P}-\text{Cd}-\text{S}$ (0.006)
- C: $\text{Li}_2\text{O}-\text{Na}-\text{Zn}-\text{P}-\text{Cd}-\text{S}$ (0.012)

D: $\text{Li}_2\text{O}-\text{Na}-\text{Zn}-\text{P}-\text{Cd}-\text{S}$ (0.018)

E: $\text{Li}_2\text{O}-\text{Na}-\text{Zn}-\text{P}-\text{Cd}-\text{S}$ (0.024)

2.2. X-ray diffraction

X-ray Diffraction Analysis was conducted to verify the amorphous state of the glass samples using a Philips Pw1373 X-ray diffractometer. The X-ray diffractometer utilized Cu radiation ($\lambda = 1.542 \text{ \AA}$) with a Ni filter, operating at 36 kV and 20 mA. The analysis was performed at a scanning rate of 2° per minute within the angular range of $4-70^\circ$.

2.3. Density measurement

The density of each glass sample was determined by applying the Archimedes principle, utilizing toluene according to the relation:

$$\rho = p_t W_{ga} / (W_{ga} - W_{at}) \quad (1)$$

Here, p_t represents the density of toluene ($p_t = 0.864 \text{ g cm}^{-3}$), while W_{ga} and W_{at} denote the weights of the glass sample in air and toluene, respectively. Additionally, the molar volume (V_m) of the samples was calculated using the formula $V_m = \sum(x_i \times m_i) / \rho$, where x_i represents the molar fraction and m_i is the molecular weight of the i th component.

2.4. Optical properties

The transmittance spectra of the glasses were recorded with a UV-VIS spectrophotometer (Unico UV-VIS double beam model 4862, USA) in the spectral range 190–1100 nm at room temperature.

Analyzing these spectra is crucial, as identifying the characteristics of absorption edges, defect-associated absorption bands, and total transmittance patterns enables the design of glass compositions to achieve the desired spectral properties. High transmittance is essential for maintaining optical clarity, enhancing its usefulness in a variety of shielding applications. Furthermore, the high light transmission rates of phosphate glass shielding materials enable direct optical inspection without compromising protection. This is particularly important in applications requiring real-time monitoring, such as during radiotherapy procedures.

Davis and Mott (Mott and Davis, 1979) hypothesized the following relationship between incoming photon energy and the optical absorption coefficient, $\alpha(\nu)$:

$$\alpha(\nu)h\nu = [B(h\nu - E_{opt.})]^q \quad (2)$$

where B is constant, $E_{opt.}$ is the optical band gap, and the exponent q can be either 2 or $1/2$ for an allowed indirect transition (AIT) and an allowed direct transition (ADT), respectively. The wavelength, λ , can be used to convert Eq. (1), as follows:

$$\alpha(\lambda) = B(hc)^{q-1} \lambda \left(\lambda^{-1} - \lambda_{opt.}^{-1} \right)^q \quad (3)$$

where h is Planck's constant, c is the light velocity and $\lambda_{opt.}$ is the wavelength related to the $E_{opt.}$. The Beer-Lambert law states that the $\alpha(\nu)$ may be found using:

$$\alpha(\nu) = \frac{2.303A}{t} \quad (4)$$

The $A(\lambda)$ can be derived by using Eqs. (3) and (4) (Alarcon et al., 2007):

$$A(\lambda) = \frac{B(hc)^{q-1}}{2.303} t \lambda \left(\lambda^{-1} - \lambda_{opt.}^{-1} \right)^q = K \lambda \left(\lambda^{-1} - \lambda_{opt.}^{-1} \right)^q \quad (5)$$

Equation (5) allows us to determine the optical band gap ($E_{opt.}$) by analyzing the absorption data (A) as a function of wavelength (λ) directly without measuring the sample thickness (t). The thickness (t) is

incorporated into the constant K ($K = [B(hc)q^{-1} t/2.303]$). The term K is dependent on q because it dictates the nature of the electronic transition (direct or indirect), and also incorporates other factors like the constant B, Planck's constant (h), and the speed of light (c). The λ_{opt} can be estimated by the extrapolation of the linear region of $(A\lambda^{-1})^q$ against (λ^{-1}) at $(A\lambda^{-1})^q = 0$.

2.5. Nuclear radiation shielding parameters

The calculation of nuclear radiation shielding parameters such as mass attenuation coefficients (MAC), half-value layer (HVL), effective atomic number (Z_{eff}), electron density (N_{eff}), mean free path (MFP), atomic cross section (ACS), electron cross section (ECS), exposure buildup factor (EBA) and energy absorption buildup factor (EABF) were performed using the user-friendly online Photon Shielding and Dosimetry (PSD) software, which offers advanced tools for precise dosimetry calculations and photon shielding analysis. This software can be accessed at <https://phy-x.net/PSD> (Sakar et al., 2020), and is used to calculate theoretical radiation shielding parameters with high efficiency and accuracy.

2.6. Mass attenuation coefficient (MAC)

The mass attenuation coefficient (MAC) is a parameter that known as the linear attenuation coefficient of materials per unit mass, reflecting their ability to interact with incident radiation. It used as a fundamental property influencing the material's capacity to absorb, scatter, and attenuate radiation. The MAC is determined by the ratio of the linear attenuation coefficient (μ) to the material density (ρ), and it plays a role in assessing the material's effectiveness in attenuating radiation across various energy levels. The following formula yields the calculation of the total mass attenuation coefficients, with the material's thickness (x) and density (ρ).

$$MAC = \frac{\mu}{\rho} = \frac{1}{\rho x} \ln \frac{I_0}{I} \quad (6)$$

where I and I_0 are the final and initial intensities of the gamma rays, respectively.

2.7. Role of Z_{eff} and N_{el} in photon interaction

The effective atomic number (Z_{eff}) and effective electron density (N_{el}) play important roles in understanding and characterizing the interaction of photons with matter, particularly in the case of radiation shielding and attenuation. The calculation methods for Z_{eff} and N_{el} are obtained through Equations (7) and (8). Z_{eff} is indicates to the effective atomic number that symbolize the quantity of electrons per atom that interact with photons. On the other hand, N_{el} refers to the density of electrons per unit mass that interact with photons to reduce their intensity as they pass through material (Gunha et al).

$$Z_{eff} = \sigma_{t,a} / \sigma_{t,el} \quad (7)$$

$$N_{el} = \mu_m / \sigma_{t,el} \quad (8)$$

where σ_{ta} is the total atomic cross-section and σ_{tel} electronic cross-section.

2.8. Half-value layer (HVL) and mean free path (MFP)

The half-value layer (HVL) can be calculated using Equation (9) as follows:

$$HVL = 0.693 / \mu \quad (9)$$

where HVL refers to the thickness of the material that reduce the radi-

ation intensity by half.

The mean free path is the average distance a photon travels before interacting within the material. The formula for calculating the mean free path (MFP) is given by the following equation:

$$MFP = 1 / \mu \quad (10)$$

where μ represents the linear attenuation coefficient.

2.9. Atomic cross section (ACS) and electron cross section (ECS)

The atomic cross section (ACS) and electron cross section (ECS) can be found from equations (11) and (12). The atomic cross section (ACS) can be calculated using the equation:

$$ACS = (\mu / \rho) \times A / N_A \quad (11)$$

where.

- ACS is the atomic cross section, μ is the linear attenuation coefficient, ρ is the density of the material, N_A is Avogadro's number, A is the atomic weight of the shielding material.

The electron cross section (ECS) can be calculated using the equation:

$$ECS = ACS \times N_{eff} \quad (12)$$

where: ECS is the electron cross section and N_{eff} is the effective electron density.

By utilizing the above equations, the atomic cross section (ACS) and electron cross section (ECS) can be calculated to assess the radiation shielding properties of materials effectively.

2.10. Exposure buildup factor (EBA) and energy absorption buildup factor (EABF)

The Exposure buildup factor (EBF) can be determined using the formula:

$$EBF = \Sigma(EBF) / N \quad (13)$$

where.

- EBF is the energy buildup factor, $\Sigma(EBF)$ represents the sum of individual energy buildup factors, and N is the total number of energy deposition events.

On the other hand, the energy absorption buildup factor (EABF) can be calculated using the equation:

$$EABF = \Sigma(EABF) / N \quad (14)$$

where.

- EABF represents the energy absorption buildup factor, $\Sigma(EABF)$ is the sum of individual energy absorption buildup factors, and N denotes the total number of energy absorption events.

2.11. Fast neutron absorption characteristics (FNAC)

Fast neutron absorption characteristic (FNAC) is defined as the ability of a material to interact with and decelerate fast neutrons. They can be determined by calculating the mass removal cross section ($\Sigma R/\rho$) for the individual elements in the material, it can be found through the following equation:

$$FNAC = \Sigma R / \rho \Sigma \rho_i \quad (15)$$

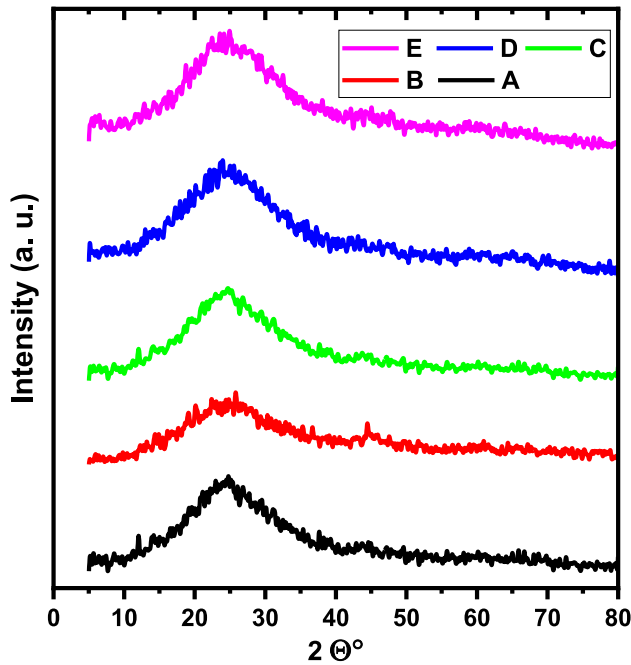


Fig. 1. XRD for prepared glass samples.

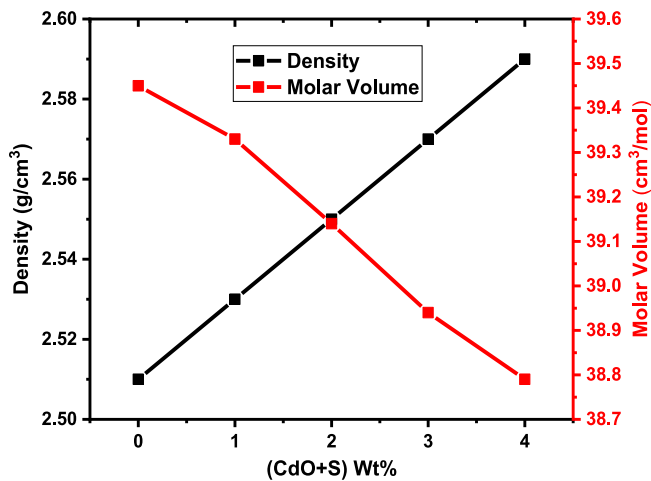


Fig. 2. Density and molar volume for prepared glass samples.

where ρ_i refers to the partial density of the oxides that make up the substance.

3. Results and discussion

3.1. The X-ray diffraction

(XRD) patterns of the prepared samples are depicted in Fig. 1, display

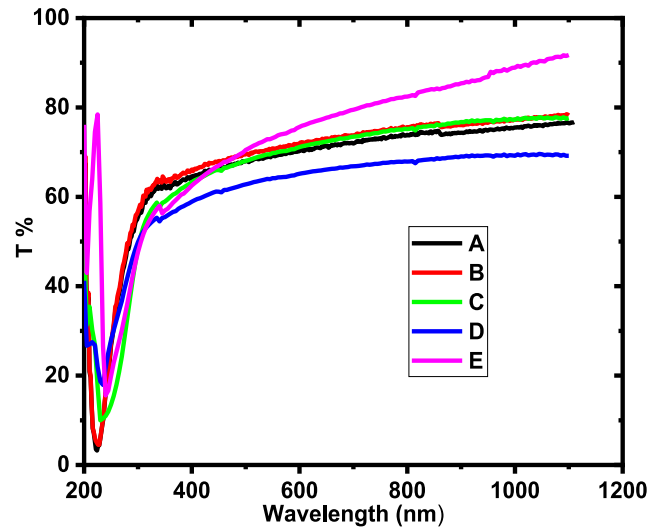


Fig. 3. Transmittance spectra of the glass samples.

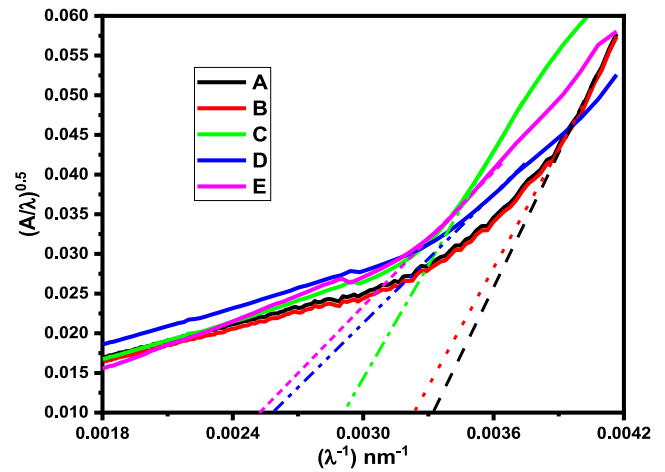


Fig. 4. Dependence of $(A/\lambda)^{0.5}$ on (λ^{-1}) for the glass samples.

all compositions. The analysis reveals that the quenched materials exhibit a distinctive feature a broad halo (or hump) at $2\theta = 24\text{--}28^\circ$. This observation refers to the fully amorphous nature of the selected materials, indicates the absence of long-range atomic arrangement and the lack of three-dimensional network periodicity in the quenched samples.

3.2. Density and molar volume

In the analysis of density and molar volume, Fig. 2 illustrates the density and molar volume variations across the prepared glass system ($5\text{Li}_2\text{O}\text{--}33\text{NaF}\text{--}7\text{ZnO}\text{--}55\text{P}_2\text{O}_5$; x (CdO + S), where $x = 0, 1, 2, 3$, and 4 wt%). Both parameters exhibit linear changes relative to the glass composition, with density increasing proportionally to the content of

Table 1

Density, molecular weights, and percentage of element oxides in the prepared glass samples.

Sample	Li ₂ O	NaF	ZnO	P ₂ O ₅	(CdO + S)	Molecular Weight of glasses (g/mol)	Density (gm/cm ³)
A	0.0500	0.3300	0.0700	0.5500	0.0000	99.1143	2.5125
B	0.0497	0.3280	0.0696	0.5466	0.0061	99.4943	2.5300
C	0.0494	0.3259	0.0691	0.5432	0.0124	99.8696	2.5517
D	0.0491	0.3239	0.0687	0.5399	0.0184	100.2402	2.5740
E	0.0488	0.3220	0.0683	0.5366	0.0243	100.6064	2.5934

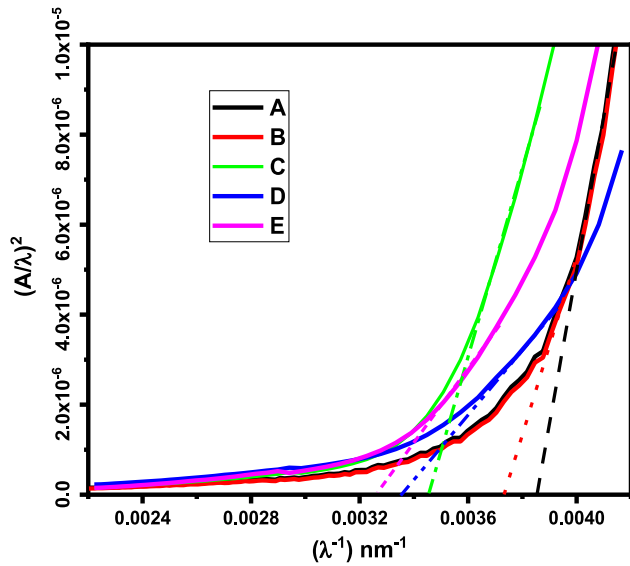


Fig. 5. Dependence of $(A/\lambda)^2$ on (λ^{-1}) for the glass samples.

Table 2

Sample code, optical band gap E_{opt}^{ASF} , and refractive index (n) for glass samples.

Sample code	E_{opt}^{ASF} (eV)		n, (ASF)	
	indirect	direct	indirect	direct
A	4.12	4.77	2.15	2.04
B	4.02	4.63	2.17	2.06
C	3.61	4.28	2.25	2.12
D	3.21	4.16	2.34	2.14
E	3.14	4.05	2.36	2.16

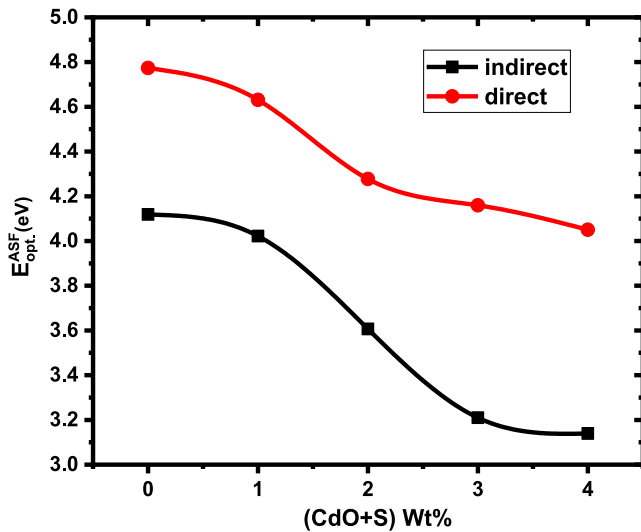


Fig. 6. Variation of $E_{opt.}$ with (CdO + S) content.

(CdO + S). This increment in density aligns with the introduction of heavier atomic masses from Cd^{2+} and S^{2-} compared to other glass components. Consequently, the packing density rises, narrowing the average interatomic spacing as (CdO + S) fills the network interstices, promoting a denser, close-packed structure. This structural adjustment results in reduced molar volume and heightened density, in agreement with prior studies (Ibrahim et al., 2023; Zahran et al., 2023; El et al.,

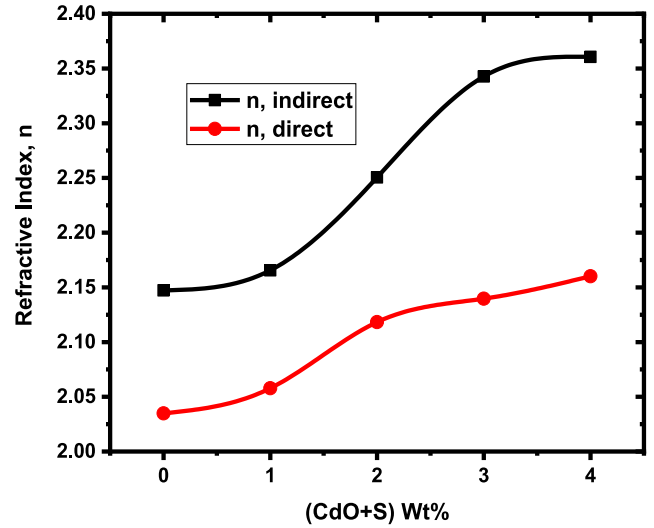


Fig. 7. Variation of n with (CdO + S) content.

2003) (Fig. 2 and Table 1).

3.3. The optical band gap and refractive index

Optical transmittance spectra in the energy range of 190–1100 nm were plotted in Fig. 3. Figs. (4) and (5) show the variation of $(A\lambda^{-1})^{1/2}$ and $(A\lambda^{-1})^2$ with (λ^{-1}) for AIT and ADT, respectively. This method used to calculate the $E_{opt.}$ is called Absorption Spectral Fitting (ASF) and the values are determined from the following relationship:

$$E_{opt.}^{ASF(ADT,AIT)} = 1239.83 \times \lambda_{opt.}^{-1} \quad (16)$$

Based on Table 2 and Fig. (6), when the (CdO + S) content increased from 0.0 to 4 wt%, the $E_{opt.}^{ASF(AIT)}$ and $E_{opt.}^{ASF(ADT)}$ of glass samples were decreased from 4.12 eV to 3.14 eV and from 4.77 eV to 4.05 eV, respectively. This decrease may be due to the Cd ions act as network modifiers in phosphate glasses. This means they disrupt the existing phosphate network structure, particularly the P–O–P bonds. This disruption leads to the formation of non-bridging oxygen (NBO) atoms and the donor center is increased by NBO atoms, thereby leading to a decrease in the $E_{opt.}$. In addition, the increase in NBO atoms significantly influences the $E_{opt.}$. NBOs have a higher polarizability compared to BO atoms. This increased polarizability leads to a reduction in the energy

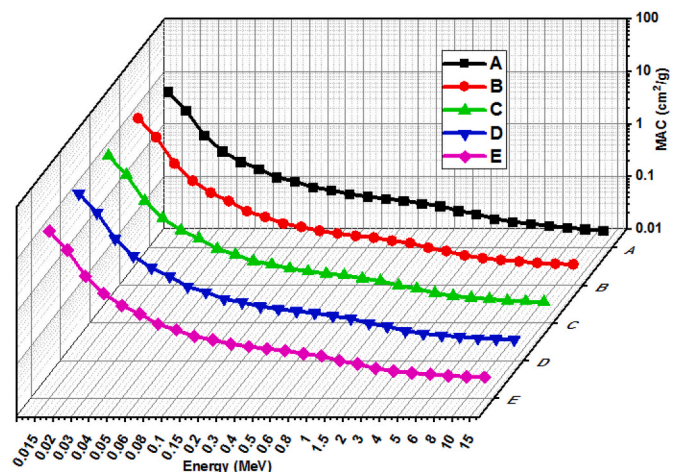


Fig. 8. Mass attenuation coeficient (MAC) for prepared glass samples.

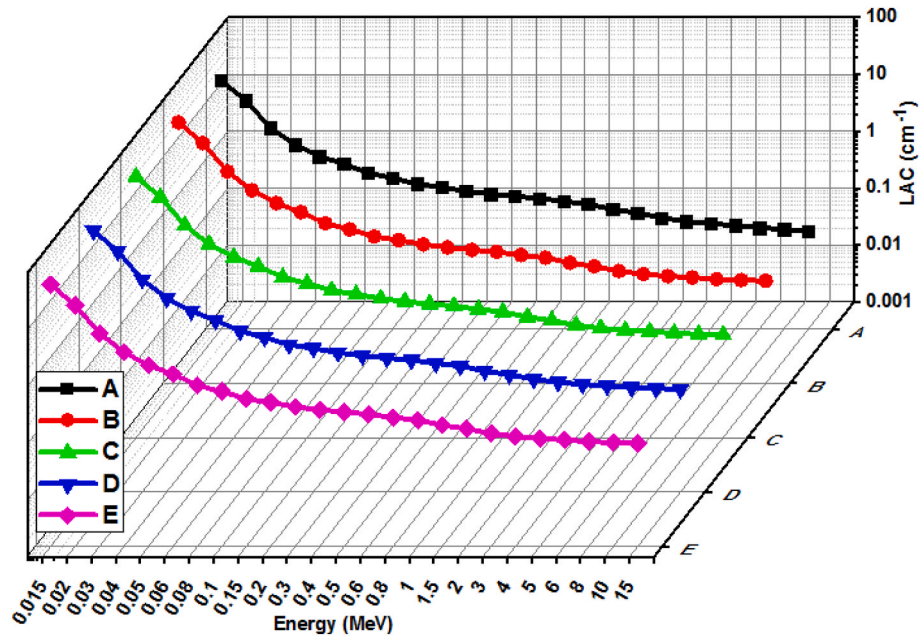


Fig. 9. Linear attenuation coefient (LAC) for prepared glass samples.

required for electronic transitions, thus decreasing the E_{opt} (Chung et al., 2015).

The refractive index, n , can be evaluated from E_{opt} values from (Dimitrov and Sakka, 1996; Algrade et al., 2017):

$$n = \left[\frac{3 - 2(E_g/20)^{1/2}}{(E_g/20)^{1/2}} \right]^{1/2} \quad (17)$$

The values of n by the $E_{opt}^{ASF(AIT)}$ and $E_{opt}^{ASF(ADT)}$ values for the samples are given in Table 2 and are depicted in Fig. 7. The increasing value of n with an increase of (CdO + S) content may be due to the increase in NBO within the glass network, and the higher polarizability of Cd^{2+} ions ($1.054 \text{ \AA}^3$) as compared to other ions in the glass (the polarizabilities of Li^+ , Na^+ , Zn^{2+} and P^{5+} ions are $0.024 \text{ \AA}^3$, $0.175 \text{ \AA}^3$, $0.283 \text{ \AA}^3$ and 0.021

$\text{\AA}^3$, respectively) (Dimitrov and Komatsu, 2010).

3.3.1. Nuclear shielding parameters

3.3.1.1. Mass attenuation coefficient. Fig. 8 shows the relationship between the mass attenuation coefficient (MAC) for glass samples at varying gamma photon energies from 0.015 KeV to 15 MeV. The MAC values were 9.37, 15.32, 15.33, 15.34 and 15.35 cm^2/g for glass A, B, C, D and E respectively. There seems to be an inverse relationship between the values of the MAC for the glass samples with photon energy. By comparing the MAC values for the different glass samples (A, B, C, D, and E) at each gamma photon energy, the trend was $A < B < C < D < E$. The sample E show higher MAC values, indicating better radiation attenuation, while sample A have lower values, suggesting reduced shielding

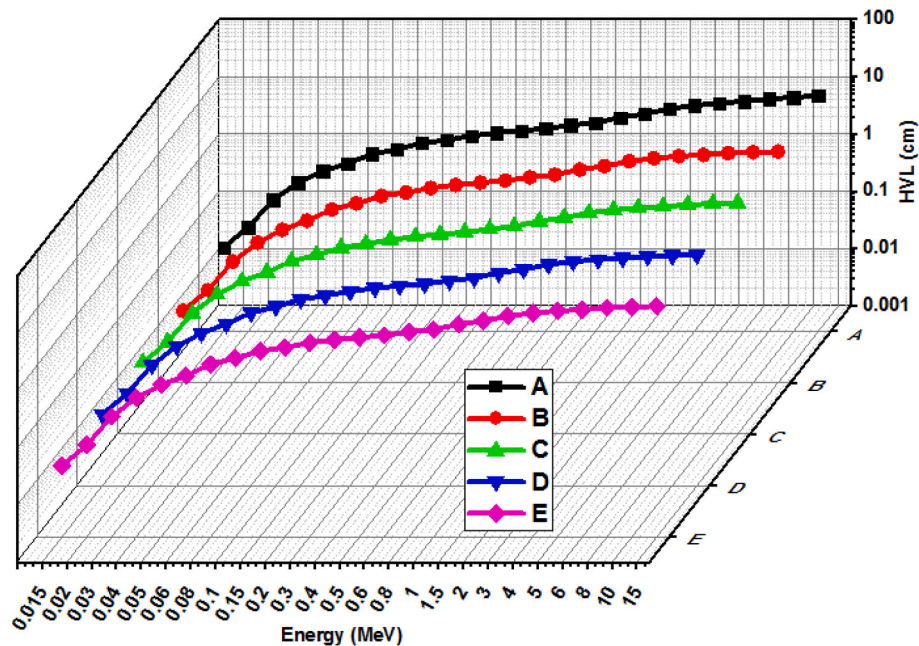


Fig. 10. Half value layer (HVL) for prepared glass samples.

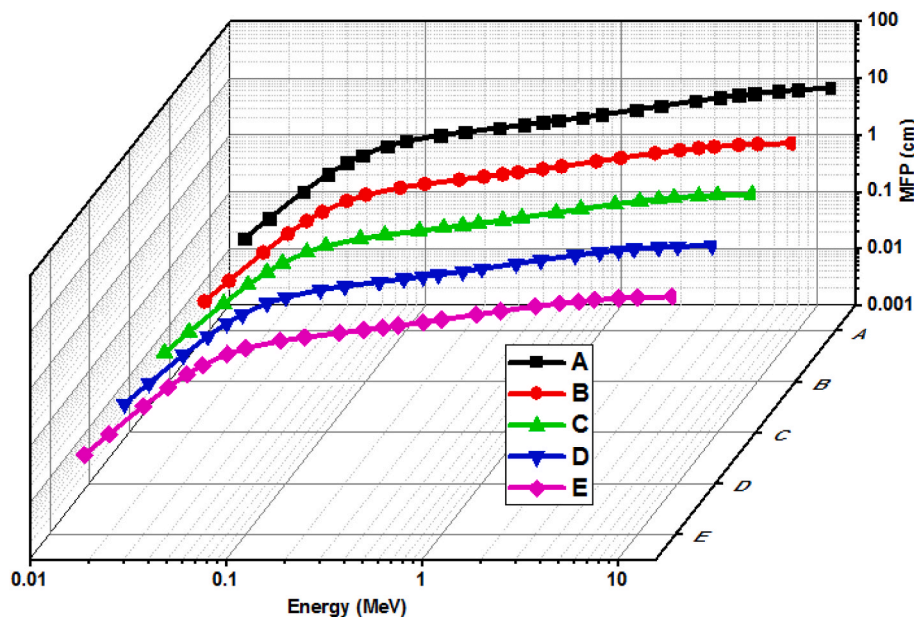


Fig. 11. Mean free path (MFP) for prepared glass samples.

effectiveness. The glass samples at 0.015 KeV show higher MAC values considered to offer better shielding efficiency at this energy. Conversely, lower MAC values were at 15 MeV ability to attenuate radiation effectively. The higher values of MAC for sample E may be contribution to the composition of the glass sample E and the density. The glass sample E have the higher weight percentage of (Cd + S) and higher density. When comparing the trends in mass attenuation coefficient (MAC) across various glass samples, a significant disparity is evident in the MAC values between samples A and B, whereas the variations in MAC values among the remaining samples are minimal.

3.3.1.2. Linear attenuation coefficient (LAC). Fig. 9 shows the LAC Trends with Gamma Photon Energies from 0.15 KeV to 15 MeV for all glass samples. The higher values were at 0.015 KeV and the lowest values were at 15 MeV. At 0.015 KeV the values of LAC were 23.56,

38.77, 39.11, 39.47 and 39.79 cm^{-1} . Higher LAC values indicate greater attenuation of gamma radiation, suggesting improved shielding effectiveness at those energy levels. It is noted that there is a significant difference in LAC values between samples A and B. This discrepancy may be due to the presence of (Cd + S) in sample B and has higher density in this samples. The minimal change between the LAC values of samples C, D, and E implies relatively consistent attenuation properties across these samples. This consistency may suggest similar material characteristics leading to comparable shielding capabilities at different photon energies. The presence and concentration of (CdO + S) in the glass compositions likely play a crucial role in determining the LAC values. Higher (CdO + S) content may lead to increased attenuation ability, resulting in higher LAC values and enhanced radiation shielding effectiveness. Sample E show the superior LAC values at studied energy ranges, which means effective shielding in comparison with other

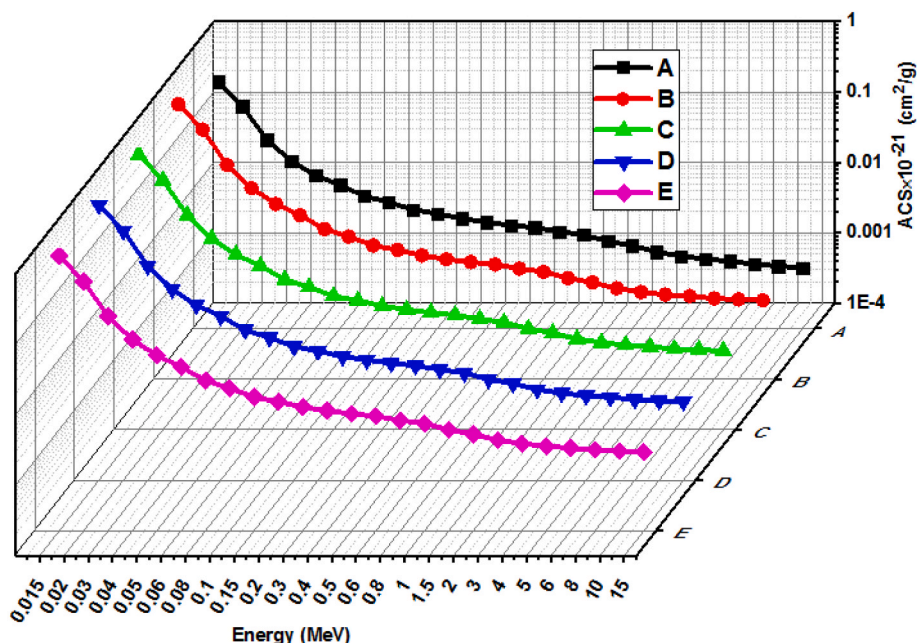


Fig. 12. Atomic cross section (ACS) for prepared glass samples.

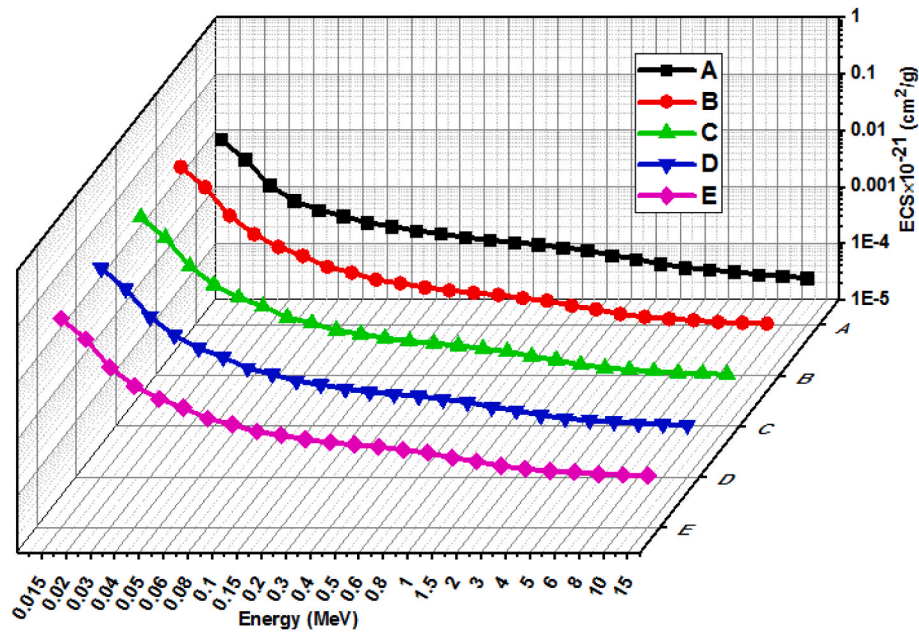


Fig. 13. Electron cross section (ECS) for prepared glass samples.

prepared glass samples.

3.3.1.3. Half value layer HVL and mean free path MFP. Fig. 10 displays the energy values corresponding to different glass samples labeled A, B, C, D, and E with the HVL values for these samples. The HVL is thickness of the material required to reduce the gamma radiation intensity by half. The higher HVL were 13.07, 10.94, 10.84, 10.75 and 10.67 cm at energy 15 KeV and lowest values were 0.029, 0.018, 0.018, 0.018 and 0.017 cm at 0.015 KeV for A, B, C, D and E glass samples respectively. By comparing the HVL values for the different glass samples at various energy levels, the trend was $A > B = C = D > E$. Sample E has lower values and sample A has higher value indicating sample E has superior shielding capabilities, requiring less material thickness for effective radiation reduction.

The mean free path MFP values indicate the average distance a gamma photon can travel through a material before being absorbed or

scattered. Fig. 11 displays the energy values versus mean free path MFP values for all prepared samples. The lower values were 0.042, 0.026, 0.026, 0.025 and 0.025 cm at 0.015 KeV and the higher values were 18.85, 15.78, 15.64, 15.51 and 15.39 cm at 15 MeV. Lower MFP values at 0.015 KeV reflect increased attenuation and absorption of gamma radiation by the materials at lower energy levels, indicating effective shielding properties. Conversely, higher MFP values at 15 MeV signify reduced interaction and greater penetration of radiation, influencing the materials' shielding efficiency at higher energy levels. The comparison of MFP values for samples E at different energy levels demonstrate superior shielding capabilities and efficient radiation absorption.

3.3.1.4. Atomic cross section ACS and electron cross section ECS. Fig. 12 displays energy values ranging from 0.015 MeV to 15 MeV beside atomic cross section for the glass samples A, B, C, D, and E. The curves illustrate a downward trend for all datasets as energy increases, reflecting a

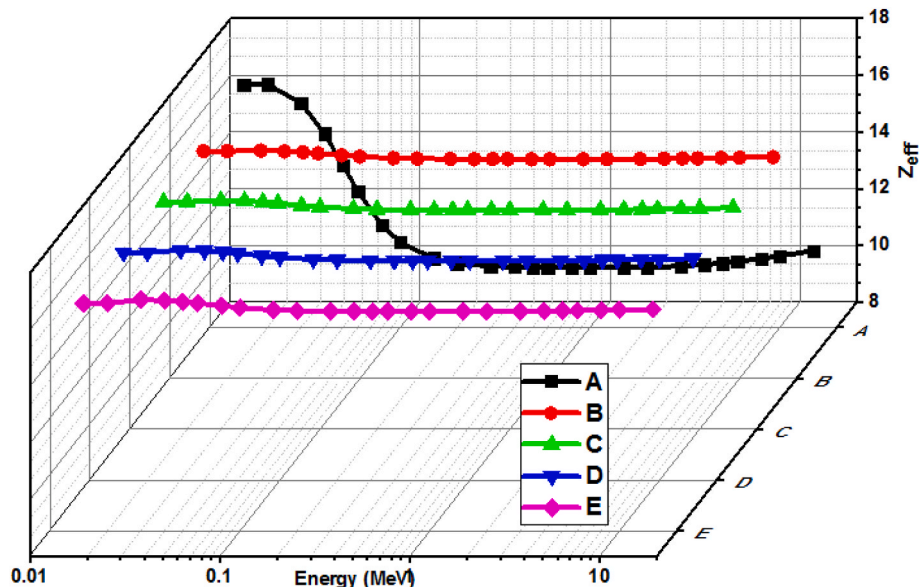
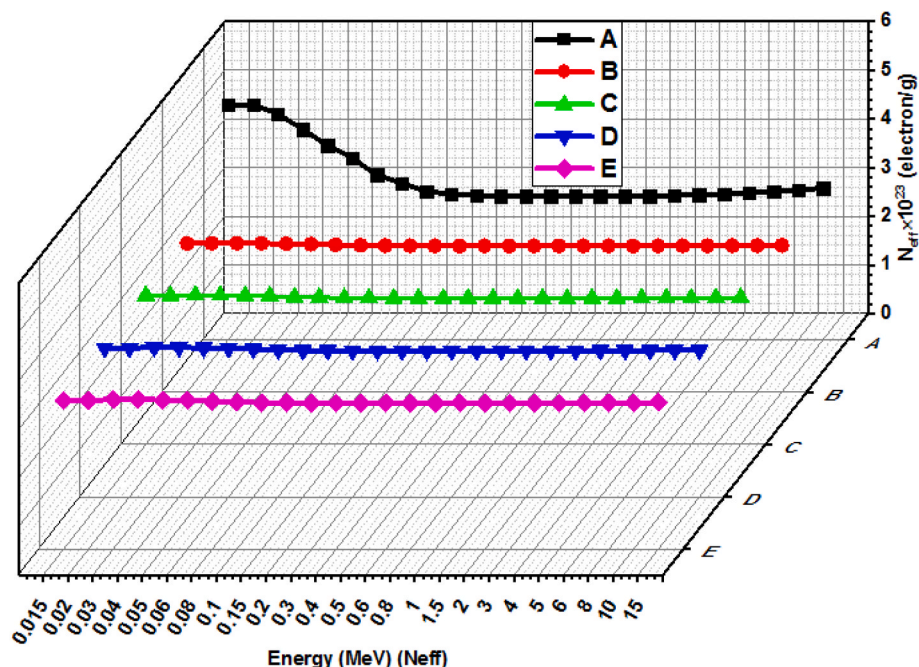


Fig. 14. Effective atomic number (Z_{eff}) for prepared glass samples.

Fig. 15. Electron density (N_e) for prepared glass samples.

decrease in atomic cross section values with higher energy levels. The decreasing trend in atomic cross section values with increasing energy levels indicates low reactivity of the materials interaction with gamma photons at higher energies. This behavior suggests that the glass samples exhibit reduced absorption or scattering of radiation as energy levels rise. The lower atomic cross section values at higher energies 15 MeV were $(0.0007, 0.0013, 0.0013, 0.0013 \text{ and } 0.0013) \times 10^{-22} \text{ cm}^2$ for A, B, C, D and E glass samples respectively, may exhibit better penetration of gamma radiation, while higher values were $(0.3215, 0.8028, 0.8032, 0.8038, \text{ and } 0.8043) \times 10^{-24} \text{ cm}^2$ at lower energies indicate enhanced interaction and absorption characteristics. Variations in atomic cross section values among the glass samples highlight differences in material composition and structure. Sample E exhibits the highest atomic cross

section (ACS), while sample A displays the lowest ACS. This difference may be attributed to the percentage of (Cd + S) content and the higher density of sample E, showing stronger shielding capabilities at specific energy ranges compared to others.

Fig. 13 illustrates the electron cross section ECS for five different glass samples as a function of energy. The ECS represents the probability of interaction between electrons and the material, with higher values indicating a greater probability of interaction. The graph depicts a trend where the electron cross section decreases as the energy of the electrons increases for all five glass samples. The decreasing trend in ECS with increasing electron energy indicates reduced interaction probability at higher energies. This suggests that electrons are less likely to interact with the materials as their energy levels rise, influencing the materials'

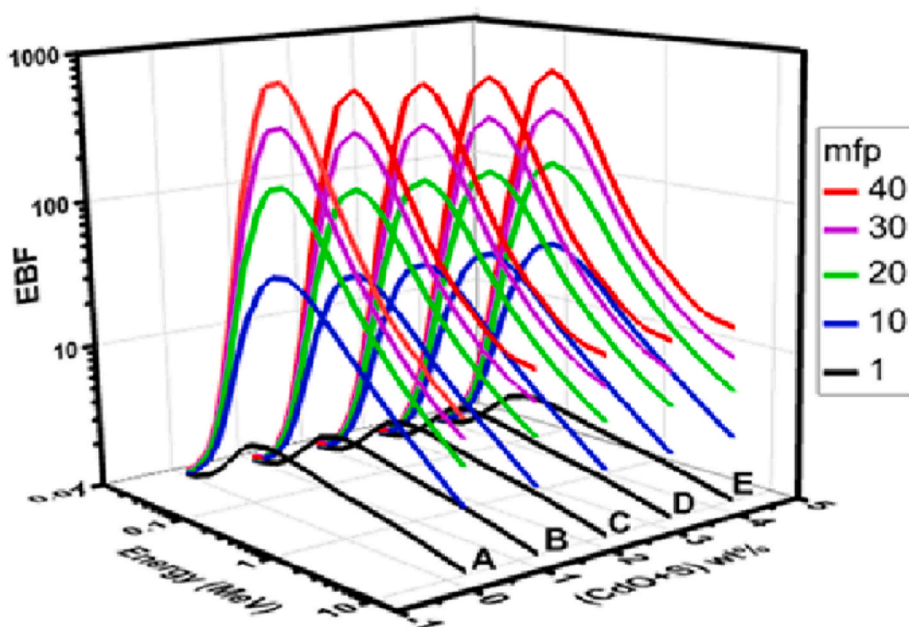


Fig. 16. Exposure build up factor (EBF) for prepared glass samples.

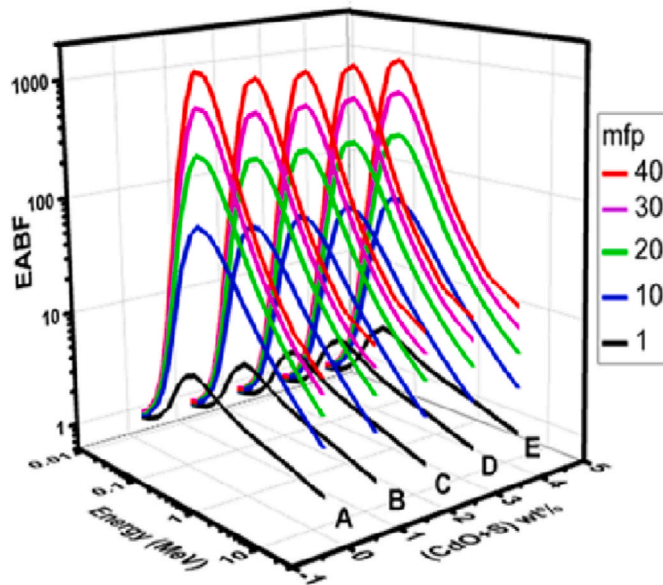


Fig. 17. Energy absorption build up factor (EABF) for prepared glass samples.

absorption and scattering properties. At energy 0.015 KeV the values of ECS were $(0.0194, 0.0501, 0.0501, 0.0501 \text{ and } 0.0502) \times 10^{-21} \text{ cm}^2/\text{g}$ for A,B,C,D and E glass samples respectively, while at energy 15 MeV the values were $(6.7659, 8.3022, 8.3027, 8.3038 \text{ and } 8.3045) \times 10^{-26} \text{ cm}^2/\text{g}$. Samples with higher electron cross sections (sample E) at certain energy levels may possess enhanced absorption capabilities, while those with lower values (sample A) might exhibit greater penetration and reduced interaction likelihood.

3.3.1.5. Effective atomic number Z_{eff} and electron density N_{eff} . Fig. 14 displays the relationship between the effective atomic number (Z_{eff}) and the energy of gamma rays in MeV. Z_{eff} represents the average atomic number of the atoms in the material and serves as a critical parameter in characterizing the interaction of gamma rays with the material. Fig. 14 illustrates how the Z_{eff} of the glass prepared samples changes as a function of the energy of the incident radiation. Indicating that the Z_{eff} decreases as the energy of the gamma rays increases. Additionally, variations in Z_{eff} values are apparent across different samples, emphasizing how the atomic composition influences the effective atomic number. The higher values of Z_{eff} were 16.57, 16.02, 16.02, 16.03 and 16.03 at 0.015 KeV and the lower values were 10.70, 15.81, 15.81, 15.82 and 15.82 at 15 MeV for A,B,C,D and E glass samples respectively. The decreasing trend in Z_{eff} with increasing gamma ray energy signifies that higher energy radiation is more likely to interact with the electrons in the material rather than the atomic nucleus. This behavior demonstrates how the sample's Z_{eff} value evolves in response to the energy level of the incident radiation. Sample with higher Z_{eff} value (E) typically have a higher concentration of heavier elements, resulting in enhanced interactions with gamma rays and increased attenuation capabilities. Additionally, Materials with higher atomic numbers exhibit higher Z_{eff} values, reflecting a greater abundance of heavier elements that contribute to a more effective interaction with gamma rays and heightened radiation absorption.

Fig. 15 allows for the analysis of how electron density changes with varying photon energy levels for each glass sample. Higher electron density values suggest a greater number of electrons present in the samples, impacting the probability of interactions with incident photons and influencing the materials' radiation attenuation properties. The higher values of N_{eff} were $(4.8323, 3.0571, 3.0577, 3.0580 \text{ and } 3.05585) \times 10^{23} \text{ electron/g}$ at 0.015 KeV and the lower values were $(3.1205, 3.0171, 3.0174, 3.0175 \text{ and } 3.0177) \times 10^{23} \text{ electron/g}$ at 15

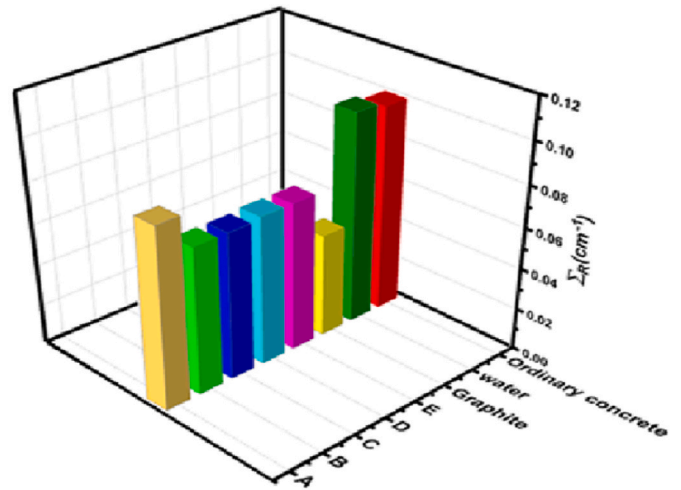


Fig. 18. Comparison of neutron removal cross section (Σ_R) for prepared glass samples with some materials.

MeV for A,B,C,D and E glass samples respectively. At all photon energy levels, sample A demonstrates a higher electron density compared to the other samples.

3.3.1.6. Exposure buildup factor (EBF) and energy absorption buildup factor (EABF). Fig. 16 presents the exposure buildup factor (EBF) values for all glass samples as a function of mean free path (MFP) of 1, 10, 20, 30, and 40. A pattern of increasing or decreasing EBF values with varying MFP distances can reveal the sample's radiation attenuation properties and energy absorption capabilities. Changes in the EBF with increasing MFP distances can indicate the sample's ability to absorb and attenuate radiation at different depths. Fig. 17 presents the energy absorption buildup factor (EABF) as a function of mean free path (1, 10, 20, 30 and 40). The energy absorption buildup factor (EABF) exhibits a similar trend to the energy buildup factor (EBF), with values increasing as the mean free path (MFP) increases. The lowest values occur at low energies, such as 0.015 KeV and high energies like 15 MeV, where the photoelectric effect and pair production dominate. Conversely, higher values are observed at intermediate energies, where Compton scattering is predominant.

3.3.1.7. Neutron removal cross section Σ_R . The neutron removal cross section values ($\Sigma_R - \text{cm}^{-1}$) for the glass samples A, B, C, D, and E are visually represented in histogram Fig. 18. The plotted values for each sample type were as follows: 0.087, 0.071, 0.071, 0.072, and 0.072 cm^{-1} , correspondingly. These values represent the effectiveness of the glass samples in removing neutrons, with higher values indicating a greater ability to absorb or attenuate neutron radiation. Among the glass samples, sample A exhibits the highest neutron removal cross section value of 0.087, indicating superior neutron shielding properties compared to samples B, C, D, and E. Samples B, C, D, and E have comparable values ranging from 0.071 to 0.072, suggesting relatively similar neutron attenuation capabilities. Sample A does not contain Cd and S, unlike the other samples. As the concentration of Cd and S in the prepared glass increases, these elements, being heavier than lithium, sodium, and phosphorus. Light elements are known for its ability to slow down fast neutrons through elastic scattering interactions, thereby thermalizing the neutrons to lower energies. As the content of Cd and S increases in the glass composition, their influence on neutron interactions may differ from that of the lighter elements, resulting in a decrease in neutron scattering efficiency. Comparing the neutron removal cross section values of the glass samples with traditional neutron shielding materials like graphite, water, and ordinary concrete can offer valuable insights. For example, graphite is known for its ability

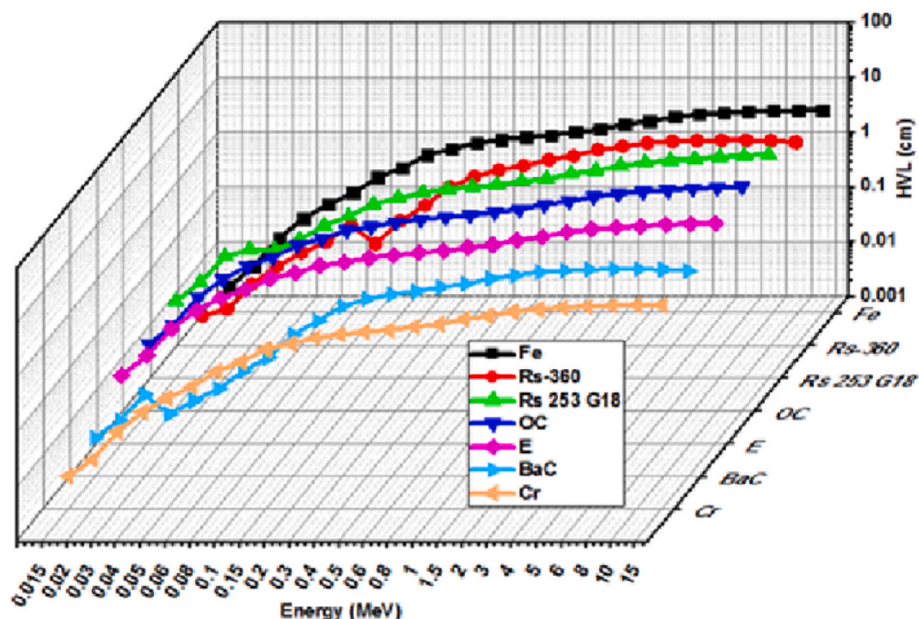


Fig. 19. Comparison of HVL for prepared samples with some comercials materials.

to slow down fast neutrons through elastic scattering interactions, thereby thermalizing the neutrons to lower energies, while water and concrete provide effective neutron moderation and shielding. Materials with higher cross section values, such as sample A, may be preferred for environments requiring enhanced neutron attenuation, while materials with lower values are suitable for different neutron shielding requirements.

3.3.1.8. Comparison of the results with other materials. Fig. 19 illustrates the relationship between the half value layer (HVL) and energy levels for various materials used as shielding against radiation at energy ranging from 0.015 to 15 MeV. Different materials, including Iron (Fe), glass Rs-360, glass Rs 253 G18, Ordinary Concrete (OC), Barium Carbonate (BaC), Chromium (Cr), and the prepared glass sample E, are depicted with distinct symbols and colors. Upon comparing the Half Value Layer (HVL) of glass sample E with traditional materials such as Iron (Fe), Rs-360, Rs 253 G18, Ordinary Concrete (OC), Barium Carbonate (BaC), and Chromium (Cr), glass sample E exhibits a lower HVL than Ordinary Concrete (OC) but higher HVL values compared to the other mentioned materials. This positioning suggests that glass sample E offers relatively efficient radiation attenuation capabilities when compared to concrete yet may have varying shielding effectiveness when compared to the other traditional materials.

4. Conclusions

The E_{opt} values of the prepared glass decrease due to the increase in NBOs and the n increases due to the increase in polarizability with the addition of cadmium ions. The calculations of various radiation shielding parameters, including MAC, LAC, HVL, MFP, Zeff, Neff, ACS, ECS, EBF, EABF, and neutron removal cross section to offers a valuable insights into the radiation attenuation capabilities of the investigated prepared glass samples. Glass sample A showed a higher neutron removal cross-section compared to samples E, B, C, and D. The comparison of glass sample E with traditional materials highlights its competitive shielding capabilities compared to ordinary concrete but also underscores variations in effectiveness when compared to other materials like iron, barium concrete, and chromium. This research contributes to advancing the understanding of prepared glass samples selection for radiation shielding purposes and informs future decisions

in nuclear safety applications.

CRediT authorship contribution statement

Emran Eisa Saleh: Conceptualization, Methodology, Software, Validation, Writing – review & editing, Supervision. **Mohammed A. Algradee:** Conceptualization, Methodology, Validation, Writing – review & editing. **Abdullah M. Qaid:** Visualization, Investigation, Resources. **Dua'a Anis Taya:** Investigation, Resources, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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